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Mobile Internet access and political outcomes: Evidence from South Africa ☆

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Highlights

- Broadband Internet shaped politics in developed economies. What about developing countries?
- In South Africa, the arrival of 3G mobile Internet between 2006 and 2016 increased voter turnout.
- It also eroded political support for the incumbent party and benefited its rivals.
- Competition at ballots intensified and the number of protests increased.
- Access to information and enhanced coordination are some potential explanations.

Abstract

I exploit the staggered introduction of 3G mobile Internet technology across South Africa to estimate its impact on political participation, electoral competition, voters' preferences, and protests. Combining granular coverage data with administrative records on municipal elections, I show that in 2016 mobile Internet availability caused a 2-percentage-point increase in voter turnout and a 3-percentage-point reduction in the vote share of the ruling party. The main opponents gained from mobile Internet arrival. The number of parties running for election and the number of protests grew. I provide suggestive evidence that both information and coordination mechanisms could explain the observed results.

Introduction

Many developing countries suffer from low political participation and low electoral competition. International observers and reporters believed that the arrival of mobile Internet could change this. However, little is known on the impact of the Internet on political participation and electoral competition, especially in developing countries. Existing literature primarily focuses on developed economies and finds that broadband Internet generally leads to lower political participation, most likely due to the increase in entertainment and crowding out of news consumption (Prior, 2005). These effects, however, could be drastically different in developing countries, where the traditional media sector is overall less competitive and the level of political information in the pre-Internet era used to be low. In this environment, Internet arrival can provide citizens with new sources of information about politics, help individuals to overcome collective action and coordination problems by reducing communication costs, and foster political change.

This paper studies whether the arrival of mobile Internet in South Africa promotes political participation and electoral competition, and affects voting behavior. Specifically, I focus on a period during which South Africa was experiencing corruption scandals and socio-economic turmoil, and look at the impact of 3G coverage on voter turnout, number of parties, and the allocation of vote shares among the major parties. In contrast to findings for developed countries (Falck et al., 2014, Gavazza et al., 2019), I find that mobile Internet in fact increased people's political participation in South Africa, and it did so immediately after its arrival.¹ I also find a positive effect on political competition, reflected in a larger number of parties running for elections. In turn, after more than two decades, the undisputed dominance of the incumbent party was damaged. Altogether, these results suggest that mobile Internet has the potential to improve the quality of the democratic process in developing countries.

Over the last 20 years, South Africa has experienced a substantial technological and political transformation. While Internet connectivity drastically improved, the incumbent party – the African National Congress – lost voters' support, and both political participation and competition grew. However, identifying the causal impact of mobile Internet access on political outcomes is challenging. Compared to the literature on radio and TV coverage,² in this case it is harder to exploit technological features of the transmitters to find an exogenous source of variation. Internet coverage is far from random, as it reflects private operators' decisions on where to install the technology (Buys et al., 2009). In this paper, I rely on a granular measure of mobile Internet coverage (1 km by 1 km on the ground) from the Global System for Mobile Communications Association (GSMA), and combine it with rich administrative electoral data. In particular, I assemble a new dataset containing geo-referenced information at the voting (sub)district-year-level on the share of area covered by 3G, political outcomes, protests, population density, luminosity as a proxy for GDP, and a variety of socio-demographic characteristics, infrastructure and geography.

To address endogeneity concerns and uncover causal relationships, I adopt two complementary strategies: a Difference-in-Differences estimator and an Instrumental Variable approach. In the first case, I exploit variation over time and space in the expansion of the 3G technology along with the change in political outcomes across approximately 35,000 partitions of voting districts between the 2006 and 2016 municipal elections. The second strategy exploits the variation in terrain ruggedness and its differential impact over time – pre- vs. post-2005, i.e. the year in which 3G became available – on the rollout of the 3G infrastructure. In both cases, I include a number of socio-economic and geographic control variables and their interaction with year dummies to flexibly account for potential confounders that may simultaneously explain 3G expansion and political outcomes.

In the most conservative specification, my findings show that a unitary increase in 3G coverage (i.e. from 0 to 100% of area covered) led to a 2-percentage-point increment in voter turnout (approximately 4% of its value in 2006) in 2011 and 2016. At the same time, mobile Internet was detrimental for the popularity of the incumbent party, whose vote share dropped by more than 3 percentage points in 2016 as a consequence of mobile Internet. By contrast, other parties gained. Moreover, as a result of 3G, competition at ballots intensified: the number of parties running for elections in more covered localities went up by 10% compared to 2006. Lastly, the number of protests against political institutions roughly tripled in 2016, suggesting that mobile Internet arrival exacerbated political discontent or facilitated mass mobilization. The results are robust to different specifications, estimators and units of observations. Various placebo exercises support the validity of the identifying assumptions.

I complement these findings with suggestive evidence on the possible mechanisms. Following Campante et al. (2018) and Manacorda and Tesei (2020), I investigate the information and coordination potential of mobile Internet for local politics.³ For voters, access to mobile Internet can significantly decrease the cost of obtaining information and, eventually, affect their voting behavior. In line with this hypothesis, I find a more detrimental impact of 3G coverage on the performance of the incumbent party in municipalities with worse administration of government finances and in localities more exposed to socio-economic turmoil. At the same time, for (potential) candidates, digital technologies could drastically reduce communication and coordination costs with voters, thus making barriers to entry into politics easier to overcome, especially for smaller groups that lack significant financial resources. In this respect, I find that mobile Internet coverage facilitated political turnover and fostered political competition by favoring the emergence of new parties. Overall, these findings suggest that the arrival of 3G technology in developing countries can provide a new source of information for voters and a cheaper mass mobilization tool for candidates.

This paper primarily contributes to the literature on political participation and competition in developing countries. Models of electoral participation suggest that more informed individuals are more likely to vote (Matsusaka, 1995, Feddersen and Pesendorfer, 1996). Other authors theoretically analyze information and political competition (Lindbeck and Weibull, 1987, Baron, 1994, Grossman and Helpman, 1996, Lohmann, 1998). However, empirical exercises are scarce, and they mainly focus on the role of awareness campaigns in local communities (Jenkins and Goetz, 1999, Goetz and Jenkins, 2001, Paul, 2002, Banerjee, 2006) and NGO activities to enhance individuals' public engagement (Boulding, 2010). This paper adds to the literature by empirically testing the determinants of political competition and participation in developing economies.

My work is also close to the literature in the political economy of media that studies the impact of broadband Internet and digital ICTs on various forms of political participation and mobilization in developed countries (Falck et al., 2014, Gavazza and Lizzeri, 2009, Campante et al., 2018) and developing economies (Manacorda and Tesei, 2020, Miner, 2015). This paper contributes to the latter, bringing novel insights on voter turnout and electoral competition, as well as on the potential underlying mechanisms. In this respect, the paper also adds to the literature on the role of information in selecting better politicians, promoting political accountability and shaping public policies (Ferraz and Finan, 2008, Besley and Burgess, 2002, Besley, 2005, Besley et al., 2005, Strömberg, 2004, Gentzkow et al., 2007, Besley and Prat, 2006, Eisensee and Strömberg, 2007, Snyder and Strömberg, 2010, Reinikka and Svensson, 2005, Olken, 2007).

Finally, this paper takes inspiration from the literature on digital media and its influence on politics (Zhuravskaya et al., 2020, Guriev et al., 2021, Bond et al., 2012, Enikolopov et al., 2020, Enikolopov et al., 2018, Allcott and Gentzkow, 2017, Chen and Yang, 2019, Petrova et al., 2021) as well as the literature on traditional media and its impact on voting behavior (DellaVigna and Kaplan, 2007, Enikolopov et al., 2011, Gentzkow et al., 2011, Gentzkow et al., 2015, Chiang and Knight, 2011, Casey, 2015, DellaVigna et al., 2014).

The remainder of this paper is structured as follows. Section 2 describes the political scenario and the media landscape. Section 3 outlines the empirical strategy, while Section 4 describes the data. Section 5 shows the main results and Section 6 investigates the mechanisms. Finally, Section 7 draws the conclusions.

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Section snippets

Political scenario

South Africa is the second largest economy and the most industrialized country of the African continent. However, poverty, unemployment and income inequality remain widespread. Government corruption, inefficient bureaucracy and political instability are among the top five challenges of doing business in the country.⁴ Local and international observers argue that the search for a policy compromise within the African National Congress ...

Determinants of 3G coverage

The rollout of the 3G infrastructure in South Africa started around 2005. At that time, existing antennas were already supporting the 2G technology, also known as Global System for Mobile (GSM) Communication, which enabled mobile phone calls and SMSs (Minges et al., 2008). 3G technology is an extension of GSM and is designed to offer faster data access speeds for mobile Internet. Three main differences exist between a 2G and a 3G mobile network infrastructure (Harris, 2011). First, more 3G ...

Sources and dataset construction

To create the dataset, I leverage rich administrative and spatial data sources. Specifically, 3G coverage data come at a very granular level (1 km by 1 km at maximum) and are collected by the GSM Association (GSMA) in partnership with Collins Bartholomew (a digital mapping provider), using official submissions from mobile network operators.²¹ Data on political outcomes at the voting district ...

Voter turnout and incumbent party vote share

Table 3 reports the results on voter turnout and the ANC share from the estimation of the Diff-in-Diff model described in Eq. (1). The coefficients of interest are those on the interaction terms (**3G** × **2016** (**2011**)). The table also reports the standalone coefficient on 3G, which captures the average difference in the dependent variables between localities with higher and lower 3G coverage at baseline (2006).

Column (3) of Table 3 shows the estimated coefficients from the model including all ...

Mechanisms

To provide suggestive evidence on the potential mechanisms, I follow Campante et al. (2018) and Manacorda and Tesei (2020) and investigate the role of information and coordination as possible drivers of the results. ...

Conclusions

In this paper, I analyzed the impact of mobile Internet coverage on political outcomes looking at the South African municipal election results over the 2000–2016 period. I mitigated concerns on potential endogeneity exploiting a newly constructed high-resolution dataset along with the staggered introduction of the 3G technology, and implemented a Difference-in-Differences and an Instrumental Variable estimation.

My findings demonstrated that the arrival of fast mobile Internet in South Africa ...

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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...Miner (2015), for instance, finds that broadband Internet led to a substantial decline in political support for the ruling coalition in Malaysia during the 2004 and 2008 elections. Similarly, Donati (2023) finds that the spread of 3G mobile Internet technology led to a decline in the vote share of the ruling party in local elections in South Africa between 2006 and 2016. This negative impact was more pronounced in corrupt localities....

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